

PROFESSIONAL SUMMARY

- Data Engineer with 5+ years of experience architecting cloud-native data platforms, building real-time and batch pipelines processing 50M+ records monthly across healthcare, audit, and energy domains.
- Owned enterprise data lake and warehouse ecosystems on AWS and Snowflake, enabling real-time analytics, regulatory reporting, and executive dashboards used by 200+ business users.
- Specialized in Spark, Kafka, Python, dbt, and distributed processing frameworks, reducing data latency by up to 60% and accelerating decision-making cycles across operational and clinical systems.
- Designed infrastructure-as-code and CI/CD-driven data deployment models using Terraform and Docker, tripling release velocity and improving pipeline reliability across multi-environment ecosystems.
- Proven leader in data governance, compliance, and observability, implementing lineage, masking, and monitoring frameworks aligned with HIPAA, GDPR, and SOX regulations.

TECHNICAL SKILLS

Languages: Python, SQL, Scala, Java, Shell
Big Data: Spark, PySpark, Kafka, Hadoop, Hive
Cloud: AWS, Azure, GCP
Warehousing: Snowflake, Redshift, BigQuery
Orchestration: Airflow, dbt, Cloud Composer
DevOps: Terraform, Docker, Jenkins, CI/CD
Visualization: Power BI, Tableau
Databases: PostgreSQL, MongoDB, Cassandra, DynamoDB

PROFESSIONAL EXPERIENCE

DATA ENGINEER

Molina Healthcare — Long Beach, CA **Jan 2025 – Present**

- Environment:** AWS Glue, Redshift, S3, Kinesis, Lambda, Kafka, Spark, Docker, Terraform, Power BI
- Architected an enterprise healthcare data lake ingesting 50M+ patient records monthly using AWS Glue and S3, enabling real-time population health analytics and improving care program targeting accuracy by 35%.
 - Led development of real-time ingestion pipelines using Kinesis and Lambda, increasing clinical data availability by 45% and accelerating intervention decisions for high-risk patient cohorts.
 - Designed Kafka-based event streaming architecture processing patient events in near real time, reducing data-to-dashboard latency to under two seconds for critical care reporting.
 - Optimized Redshift warehouse performance through partitioning and query tuning, cutting enterprise dashboard latency by 60% and improving analytics productivity across reporting teams.
 - Built 25+ self-service Power BI dashboards used by 200+ stakeholders, reducing ad-hoc reporting requests by 50% and enabling leadership-driven operational planning.
 - Implemented Python-driven data quality frameworks detecting 250+ anomalies monthly, strengthening audit readiness and ensuring HIPAA-compliant data integrity.
 - Standardized Spark job deployments using Docker and Terraform-based CI/CD pipelines, tripling release frequency while reducing production failures by 40%.
 - Launched observability framework leveraging CloudWatch and Athena monitoring, reducing pipeline downtime by 35% and improving SLA adherence.

DATA ENGINEER

KPMG India **Jul 2021 – Dec 2023**

Environment: Snowflake, AWS Glue, Python, dbt, Terraform, Tableau

- Owned migration of 30+ legacy ETL pipelines into AWS Glue and Snowflake, processing 8TB+ audit datasets and reducing infrastructure costs by 25%.
- Engineered dbt transformation frameworks with automated testing and snapshotting, standardizing enterprise KPIs and reducing cross-team development cycles by 30%.
- Built Terraform-based infrastructure automation and CI/CD deployment pipelines, improving data release velocity by 40% while minimizing manual configuration errors.
- Designed ML-ready feature engineering pipelines supporting audit risk modeling, accelerating experiment deployment timelines by 30%.
- Developed reusable Redshift and Snowflake data marts enabling self-service analytics, reducing reporting turnaround time by 50%.
- Instituted enterprise data lineage tracking using dbt documentation and ER modeling, ensuring full audit traceability and regulatory compliance.
- Streamlined batch ingestion workflows using Python automation, reducing reconciliation effort by 35% across financial datasets.
- Partnered with audit and compliance stakeholders to align data validation frameworks with SOX regulatory reporting requirements.

DATA ANALYST

Adani Group — India

Apr 2019 – Jun 2021

Environment: Python, PostgreSQL, Power BI, Excel

- Centralized operational energy data from 50+ industrial plants into PostgreSQL, establishing enterprise reporting infrastructure supporting three business units.
- Analyzed 1M+ consumption records using Python and NumPy, identifying seasonal inefficiencies that informed energy optimization strategies.
- Built executive Power BI dashboards tracking plant KPIs, enabling leadership to monitor performance in real time.
- Applied K-Means clustering to segment facilities by usage behavior, driving targeted conservation initiatives that reduce energy waste.
- Designed predictive forecasting dashboards integrating weather and production inputs, enabling proactive procurement planning.
- Implemented rule-based validation across 500K+ readings, improving forecasting model accuracy and reducing reporting errors.
- Partnered with engineering and finance leadership to translate analytics into operational cost-saving initiatives.
- Automated Excel-based reporting workflows, reducing manual effort by 40% across monthly performance reviews.

PROJECTS

IoT Fire & Smoke Detection System

- Designed an IoT-based hazard detection platform integrating MQ sensors and NodeMCU devices for real-time environmental monitoring.
- Built MQTT-based telemetry pipelines transmitting continuous gas readings to centralized monitoring systems.
- Developed an automated Telegram alerting framework notifying users when gas thresholds exceeded safety levels.
- Engineered Python-based anomaly detection scripts reducing false-positive alerts during field testing.

Climate Change & Bird Migration Analytics (Capstone)

- Engineered multi-source climate and migration datasets integrating NOAA and MODIS data for predictive modeling.
- Built machine learning models (Random Forest, Gradient Boosting) predicting migration arrival shifts with high accuracy.
- Created geospatial dashboards mapping habitat loss and migration hotspots using GeoPandas and Plotly.
- Delivered end-to-end analytics platform integrating predictive modeling, spatial mapping, and trend analysis.

EDUCATION

Master's in Information Technology

Kennesaw State University, Georgia, USA